

Syllabus for

Academic year 2020/2021 >

TMA970 - Introductory mathematical analysis

Inledande matematisk analys

Syllabus adopted 2019-02-14 by Head of Programme (or corresponding)

Owner: [TKTFY](#)

6,0 Credits

Grading: TH - Pass with distinction (5), Pass with credit (4), Pass (3), Fail

Education cycle: First-cycle

Major subject: Mathematics, Engineering Physics

Department: 11 - MATHEMATICAL SCIENCES

Teaching language: Swedish

Application code: 57136

Open for exchange students: No

Only students with the course round in the programme plan

Credit distribution

Module	Sp1	Sp2	Sp3	Sp4	Summer course 	No Sp	Examination dates 
0197 Examination 6,0 c Grading: TH	6,0 c						Wed 28/10-2020 am J  Thu 07/01-2021 pm J, Fri 27/08-2021 am J

In programs

[TKTEM ENGINEERING MATHEMATICS, Year 1 \(compulsory\)](#)

[TKTFY ENGINEERING PHYSICS, Year 1 \(compulsory\)](#)

Examiner:

[Jana Madjarova](#)

 [Go to course homepage](#)

Eligibility

General entry requirements for bachelor's level (first cycle)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Specific entry requirements

The same as for the programme that owns the course.

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Course specific prerequisites

Studies at the university begin with a two weeks non-compulsory introduction which is a repetition of the main features of high school mathematics. The introduction (about 30 hours) consists of:

- information and diagnostic test
- algebraic expressions
- trigonometry
- analytic geometry
- functions.

Literature: R. Pettersson: Förberedande kurs i matematik vid CTH

Aim

The course provides basic knowledge of mathematical analysis, which is necessary for most courses to follow at the Engineering Physics programme.

Learning outcomes (after completion of the course the student should be able to)

- understand the fundamental notions and definitions of mathematical analysis;
- prove the most fundamental theorems within analysis of functions in one real variable;
- use mathematical induction to prove identities and inequalities;
- rewrite expressions which contain logarithms and the inverse of the trigonometric functions;

- use a combination of standard limits in order to find other limits;
- analyze functions in order to draw their graphs;
- use standard methods to find the antiderivatives of some elementary functions;
- use the main theorem of analysis to compute Riemann integrals;
- apply Riemann integration on length, area and volume computation;
- use comparison methods to determine convergence/divergence of improper integrals;
- give proves of his/her own;
- solve problems combining two or more of the above abilities.

Content

Elementary set theory and logics. Proof by induction. Real numbers, absolute value, inequalities, Dedekind's intersection theorem for intervals, supremum / infimum. Functions, inverse functions. Exponential, power and logarithmic functions. Trigonometric functions and their inverse functions. Limits, continuity. Derivatives, derivation rules, differentials, the mean value theorem of differential calculus. Construction of curves, asymptotics, local extrema, maximal / minimal values, convex functions. Numerical solution of equations, iterative methods, method of Newton - Raphson (MATLAB, optional). Indefinite integrals, integration by parts, change of variables. The Riemann integral. Riemann sums and integration of continuous functions. The main theorem of integral calculus. The main theorem of analysis. The mean value theorem of integral calculus. Improper integrals. Numerical integration (MATLAB). Applications of integrals (volumes, length of curves etc).

Organisation

Lectures, exercises, Matlab .

Literature

Litteratur: A. Persson, L.C. Böiers: Analys i en variabel (Studentlitteratur, Lund)

Exercises: Analys i en variabel. Studentlitteratur.

Examination including compulsory elements

Written paper which combines theory and problem solving.